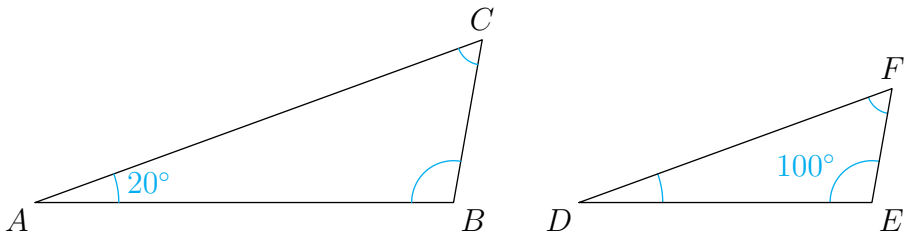


Exercises for Chapter 0

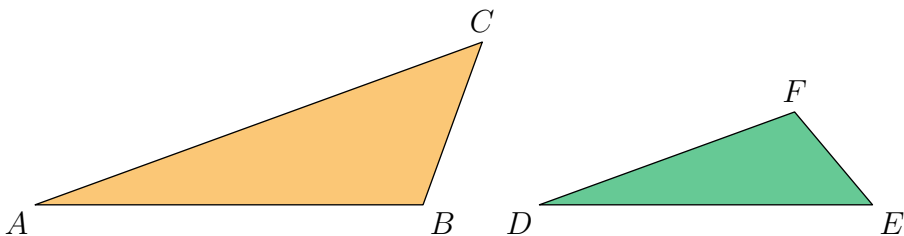
0.1.1

The triangles are similar. Find the value of $\angle ACB$.



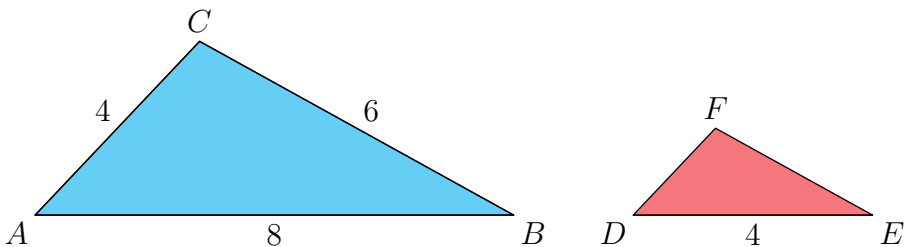
0.1.2

The triangles are similar. Find the three pairs of corresponding sides.



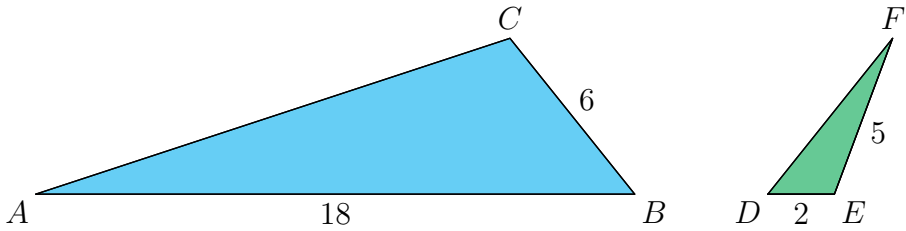
0.1.3

The triangles are similar. Find the length of EF and the length of DF .



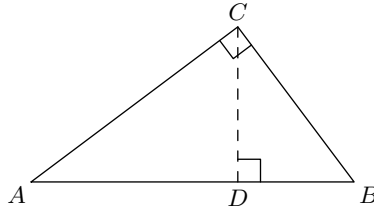
0.1.4

The triangles are similar. Find the length of AC and the length of DF .



0.1.5

Find all the similar triangles defined by A , B , C og D .



0.1.6

$\triangle ABC$ and $\triangle DEF$ are similar.

- What is the ratio of the area of $\triangle DEF$ to the area of $\triangle ABC$ if $h_1 = 2$ and $h_2 = 6$?
- Given a number a . What is the ratio of the area of $\triangle DEF$ to the area of $\triangle ABC$ if $h_2 = ah_1$?

0.1.7

En kjegle has radius 10 and height 4.

- a) Find the surface area of the cone.
- b) Find the volume of the cone.

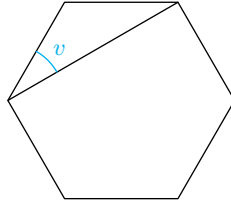
0.1.8

- a) Two spheres have radii 2 and 6, respectively. What is the ratio of the volume of the larger sphere to the volume of the smaller sphere?
- b) A sphere has radius r , and another sphere has radius ar , where $a > 1$. What is the ratio of the volume of the larger sphere to the volume of the smaller sphere?

Ponder 1

(GV21D1)

The polygon is a regular¹ hexagon. Find the value of v .

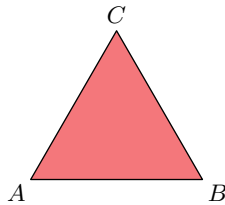


Ponder 2

Given an isosceles triangle $\triangle ABC$, with $AC = BC$. Prove that² the bisector of $\angle ACB$ is the perpendicular bisector of AB .

Ponder 3

$\triangle ABC$ is equilateral and have sides of length s .



- a) Prove that in a triangle with angles 30° , 60° , 90° , the hypotenuse is twice as long as the shortest leg.
- b) Prove that the height of $\triangle ABC$ is $\frac{\sqrt{3}}{2}s$.

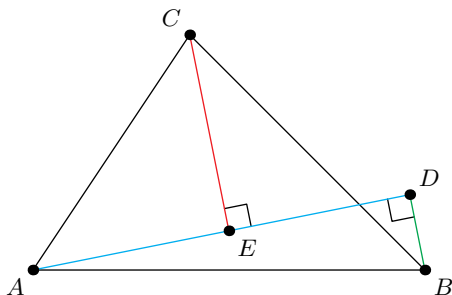
¹Regular polygons have sides of equal length.

²The definition of the bisector of an angle and the perpendicular bisector of a segment is found in [TM1](#).

Ponder 4

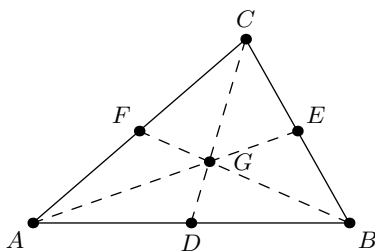
Prove that the double area of $\triangle ABC$ can be written as

$$AE \cdot BD + CE \cdot AD$$



Ponder 5

A **median** of a triangle is a segment joining a vertex to midpoint of the opposite side.



Given $\triangle ABC$ with medians AE , BF and CD .

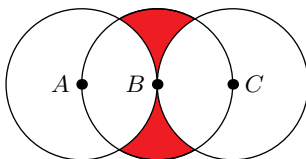
- Prove that AE , BF and CD intersect at exactly one point (G in the figure).
- Prove that

$$\frac{GC}{DG} = \frac{GB}{FG} = \frac{GA}{EG} = 2$$

Note: Part b) is probably easier than part a).

Ponder 6

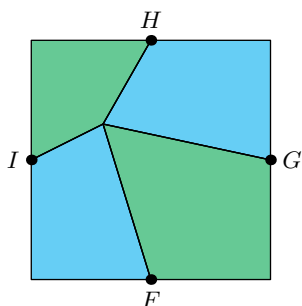
The three circles have radius 2, and A , B and C lie on the same line. Find the area of the red surface.



Hint: You could get use of the fact that a sector with angle v makes up $\frac{v}{360^\circ}$ of the area of a circle with equal radius.

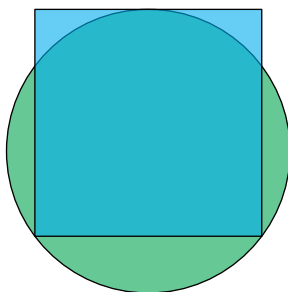
Ponder 7

The colored surfaces makes up a square, and F , G , H and I are, respectively, the square's midpoints. Prove that the area of the blue surface equals the area of the green surface.



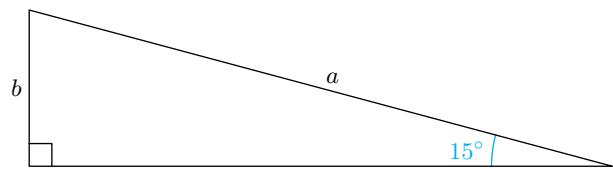
Ponder 8

The square has sides of length 4. Find the radius of the circle.



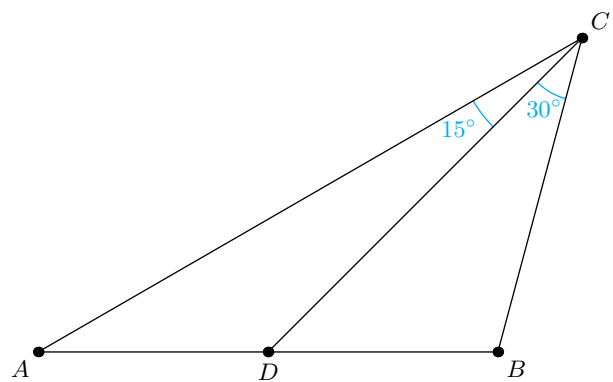
Ponder 9

a) Prove that $\frac{a}{b} = \sqrt{2} + \sqrt{6}$.



Note: To solve this problem you may (but far from necessarily) need the *abc*-formula found in [TM1](#).

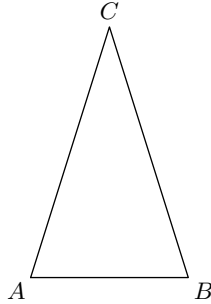
b) $AD = BC$. Find the value of $\angle A$.



Ponder 10

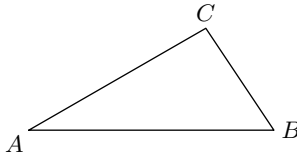
Note: This exercise deals with results that at first seems obvious, but turns out rather hard to prove.

- a) Prove that if $AC = BC$, then $\angle A = \angle B$.

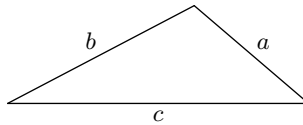


Note: Earlier on, we have declared that an isosceles triangle has two equal angles, but strictly we have to prove it.

- b) Prove that if $AC > BC$, then $\angle B > \angle C$.



- c) Given $\triangle ABC$, where AB is the longest side. Prove that when AB is the base, the height lies inside the triangle.
- d) In the figure below, c is the longest side of the triangle.



Prove that

$$c > a + b \quad , \quad b + c > a \quad , \quad a + c > b$$

Note: These three inequalities are called the **triangle inequality**.